

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Minor Project – Zeroth Review

Abstract

Urban traffic congestion has become one of the most critical challenges in modern cities, resulting in excessive delays, fuel wastage, and increased emissions. Conventional traffic signal systems operate on fixed or pre-timed cycles, which fail to adapt to fluctuating real-time traffic conditions. This project proposes an AI-powered dynamic traffic signal optimization system using Reinforcement Learning (RL) to intelligently manage urban intersections. The objective is to enable adaptive control of signal phases based on live traffic flow, thereby minimizing waiting time, congestion, and fuel consumption.

The proposed system utilizes the SUMO (Simulation of Urban Mobility) environment integrated with Python and PyTorch for training and testing the RL agent. The agent observes real-time parameters such as lane density, queue length, and vehicle waiting time as its state inputs, and decides the optimal signal phase and duration as actions. The reward function is designed to penalize congestion and reward smooth flow, encouraging the agent to discover efficient signal patterns autonomously. Over multiple episodes of training, the agent learns policies that improve average traffic throughput and reduce intersection delays.

Expected outcomes include significant reductions in average waiting time, queue length, and CO₂ emissions, along with enhanced adaptability across varying traffic conditions. By leveraging reinforcement learning for urban traffic management, this project contributes to Sustainable Development Goal 11 (Sustainable Cities and Communities) by promoting intelligent, data-driven transportation systems that ensure efficient mobility and reduced environmental impact. The system lays the groundwork for scalable AI-based smart city infrastructure capable of supporting real-world deployment in the future.

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